

A • MAGNETS — Introduction & Classification

Magnet	Object that attracts materials like iron, cobalt and nickel — property called MAGNETISM
Magnetism	Branch of physics that deals with the properties of magnets — can be natural or induced
Magnetite	Strongest natural magnet — iron oxide ore (Fe_3O_4) — also called Lodestone — found in sandy deposits
Origin of name	Magnets discovered in Magnesia (region of Asia Minor) — named after the place; also named after shepherd Magnus whose iron-tipped staff stuck to rocks
History	Chinese knew magnets before 200 BC — used magnetic compass for navigation from 1200 AD onward

Natural vs Artificial Magnets — Comparison

Feature	Natural Magnets	Artificial Magnets
Source	Found in nature (sandy deposits of earth)	Made in laboratory or factory by humans — also called man-made magnets
Shape	Irregular shapes and dimensions	Made in various specific shapes and dimensions
Strength	Well determined; difficult to change	Can be made with required and specific strength; usually STRONGER than natural
Durability	Long lasting — permanent	Time bound — may lose properties over time
Usage	Limited practical use	Vast usage in daily life
Examples	Magnetite (Fe_3O_4), Pyrrhotite (FeS), Ferrite, Coulumbite	Bar magnet, Horseshoe, U-shaped, Ring, Disc, Cylindrical, Electromagnets

Natural Magnetic Ores (Iron Ores)

Magnetite (Fe_3O_4)	72.4% iron — highest iron content — strongest magnetic property — used as natural magnet
Hematite (Fe_2O_3)	69% iron — common iron ore — less magnetic than magnetite

Siderite (FeCO₃)

48.2% iron — weakest of the three — least magnetic

Artificial Magnet Types — Shapes

- Bar magnet — rectangular bar shape; most common type used in labs
- U-shaped / Horseshoe magnet — U-shaped; both poles face same direction; strong pull
- Ring magnet — circular ring; poles on flat faces; used in speakers
- Cylindrical / Disc magnets — used in electronic devices
- Needle magnet — thin, pointed; used in compass
- Electromagnet — coil of wire wound around iron core; magnetic only when current flows
- Made from: Iron, Nickel, Cobalt, Steel alloys; Neodymium alloy (strongest); Samarium alloy; ALNICO (Al+Ni+Co)

B · PROPERTIES OF MAGNETS — Three Key Properties

■ Three properties of magnets: (1) Attractive Property (2) Repulsive Property (3) Directive Property

1. Attractive Property

- Magnets attract **MAGNETIC MATERIALS** — substances containing iron, cobalt, nickel
- Attractive force is **MAXIMUM** at the **POLES** (ends of magnet) — minimum at the centre
- Iron filings experiment: filings concentrate at poles, sparse in middle — proves poles have maximum attraction
- Magnetic substances (attracted): Iron, Cobalt, Nickel, Steel and their alloys
- Non-magnetic substances (not attracted): Paper, Plastic, Wood, Aluminium, Copper, Glass, Rubber
- Magnetic force acts through non-magnetic materials (paper, glass, wood) without them being affected

2. Repulsive Property

- **Like poles REPEL** — N-N repel; S-S repel
- **Unlike poles ATTRACT** — N-S attract; S-N attract
- Repulsion is the **SUREST** test to identify a magnet (attraction alone can be due to induction)
- Force of attraction/repulsion is **GREATER** when charged objects are **CLOSER** to each other
- Breaking a bar magnet into pieces: **EACH** piece becomes a complete magnet with its own N and S pole — magnetic poles **ALWAYS** exist in pairs
- Bar magnet broken into 2 pieces → 2 magnets | into 4 pieces → 4 magnets — each always has both poles

- Splitting magnet vertically → length changes | splitting horizontally → length unchanged — both reduce strength

3. Directive Property

- A freely suspended bar magnet ALWAYS aligns itself in geographic NORTH-SOUTH direction
- North pole of magnet points toward geographic NORTH; South pole toward geographic SOUTH
- Used in making magnetic compass — needle is a tiny pivoted magnet that rotates in horizontal plane
- This directive property is the basis of navigation using compass
- Magnetic Meridian: line passing through north and south magnetic poles at a given location

C · MAGNETIC FIELD & MAGNETIC FIELD LINES

Magnetic Field	Space around a magnet where its magnetic effect or influence is observed — measured in tesla or gauss
Unit	1 tesla (T) = 10,000 gauss — SI unit is tesla; gauss is CGS unit
Magnetic Field Lines	Curved lines drawn around a magnet showing direction of magnetic force — traced using compass needle
Plotting Compass	Tiny pivoted magnet in a case — also called compass needle — can rotate freely in horizontal plane

Properties of Magnetic Field Lines

- Field lines emerge from NORTH pole and enter SOUTH pole OUTSIDE the magnet
- Field lines travel from SOUTH pole to NORTH pole INSIDE the magnet
- Field lines are CLOSED CURVES — they form complete loops
- Field lines NEVER cross or intersect each other
- Field lines are CLOSER together where field is STRONGER (near poles)
- Field lines are FARTHER apart where field is WEAKER (away from poles, middle)
- Direction of field line at any point = direction a free North pole would move at that point
- At one point between two like poles facing each other, field lines cancel → NEUTRAL POINT (no magnetic force)

Uniform vs Non-Uniform Magnetic Field

	Uniform Magnetic Field	Non-Uniform Magnetic Field
Field lines	Parallel, equally spaced, same direction	Not parallel; varying distance between lines

	Uniform Magnetic Field	Non-Uniform Magnetic Field
Strength	Same magnitude at all points in the region	Different magnitude at different points
Example	Between poles of a very large horseshoe magnet at centre	Near the poles of a bar magnet

D · MAGNETIC MATERIALS — Diamagnetic, Paramagnetic & Ferromagnetic

Type	Alignment in Uniform Field	Movement in Non-Uniform Field	Magnetisation	Effect of Temperature	Examples
Diamagnetic	PERPENDICULAR to field direction	Moves from STRONGER to WEAKER region	Opposite to the applied field	NOT affected by temperature	Bismuth, Copper, Mercury, Gold, Water, Alcohol, Air, Hydrogen
Paramagnetic	PARALLEL to field direction	Moves from WEAKER to STRONGER region	In the SAME direction as field	Affected by temperature	Aluminium, Platinum, Chromium, Oxygen, Manganese, Nickel/Iron salt solutions
Ferromagnetic	STRONGLY PARALLEL to field direction	Moves QUICKLY from weaker to STRONGER region	STRONGLY in the direction of field	Affected by temperature — become paramagnetic on heating (Curie temperature)	Iron, Cobalt, Nickel, Steel and their alloys

■ **Curie Temperature:** The temperature at which a ferromagnetic material becomes paramagnetic. Named after Pierre Curie.

Magnetically Hard vs Soft Materials

Type	Magnetisation	Retains Magnetism?	Used For	Examples
Magnetically Hard	Requires STRONG magnetic field to magnetise — difficult	YES — retains magnetism (permanent magnets)	Making permanent magnets	Hardened steel, ALNICO alloy, Neodymium alloy

Type	Magnetisation	Retains Magnetism?	Used For	Examples
Magnetically Soft	Easily magnetised — requires weak field	NO — loses magnetism when field removed (temporary magnets)	Making electromagnets, transformer cores	Soft iron, Mumetal

E · PERMANENT & TEMPORARY MAGNETS — Methods & Storage

Permanent Magnets

Definition	Artificial magnets that RETAIN magnetic property even without external magnetic field
Material	Made from hardened steel, ALNICO (Al+Ni+Co alloy), Neodymium alloy, Samarium alloy
Strongest magnet	Neodymium magnets — strongest and most powerful artificial magnets on Earth
ALNICO	Alloy of Aluminium, Nickel and Cobalt — most commonly used for permanent magnets
Examples	Refrigerator door seal, speakers, fridge magnets, magnetic compass, credit/debit card strip

Temporary Magnets (Electromagnets)

Definition	Magnets produced with help of EXTERNAL magnetic field — lose property when field removed
Material	Made from SOFT IRON — easily magnetised and demagnetised
How made	Coil of insulated wire wound around soft iron core — becomes magnet when current flows; loses magnetism when current stops
Examples	Magnets in electric bells, cranes, MRI machines, electric motors

Methods of Magnetisation

Single touch method	Stroke the magnetic material with ONE pole of a magnet in ONE direction only — repeat 20-30 times
Double touch method	Stroke with both poles simultaneously from centre to ends
Electrical method	Pass electric current through a coil wound around the material — most common industrial method

Hammering method

Strike unmagnetised iron in N-S direction — weak magnetisation

Ways Magnets LOSE their Properties

- HEATING a magnet to high temperature — thermal energy disrupts magnetic alignment
- Continuous HAMMERING of the magnetic substance
- DROPPING the magnet from a height — mechanical shock
- Placing magnet IDLE for a very long time
- Passing VARIABLE CURRENT in a coil that encloses the magnet (AC demagnetisation)
- IMPROPER STORAGE of the magnet
- Placing magnet near CELLPHONE, COMPUTER, DVDs — their magnetic/electric fields demagnetise

Storage of Magnets

- Bar magnets: stored in PAIRS with unlike poles on same side; separated by a piece of wood; soft iron pieces placed across ends as keepers
- Horseshoe magnets: single piece of soft iron (keeper) placed across the poles
- Keep away from electronic devices, cellphones, computers, DVDs

F · MAGNETIC COMPASS — Discovery, Types & Working

Compass	Instrument to find directions — uses directive property of magnets — contains a magnetic needle
Magnetic Needle	Tiny pivoted magnet (pointer) that can rotate freely in horizontal plane — points N-S
Mariner's Compass	Compass used by sailors/seamen to find directions during sea voyages — oldest navigation tool
Chinese discovery	Chinese discovered that suspended lode stone (magnetite) stops in N-S direction — ~800 years ago used in boats during storms
Compass use	Used in ships, airplanes, military, mountaineers to find directions in unknown places

Making a Homemade Compass

- Magnetise a steel needle by stroking with a magnet (single touch method) 20-30 times
- Insert the magnetised needle into two styrofoam (thermocool) balls or a dry leaf/cork
- Float in a bowl of water — needle rotates and rests in N-S direction
- Marked end of compass needle = North Pole of the magnet inside

G · EARTH'S MAGNETISM

Earth as Magnet	Earth behaves as a huge magnetic dipole — imagined as a giant bar magnet inside
Geographic North Pole	Earth's SOUTH magnetic pole is located near geographic North Pole (attracts magnet's North)
Geographic South Pole	Earth's NORTH magnetic pole is located near geographic South Pole (attracts magnet's South)
Magnetic Axis	Line joining Earth's magnetic poles — inclined at 10° – 15° to geographical axis (rotation axis)
Magnetic Declination	Angle between geographic north and magnetic north at any point on Earth's surface
Earth's field strength	Ranges from 25 to 65 micro tesla at Earth's surface
Earth's core	Liquid outer core (molten metallic fluid, radius ~ 3500 km) believed to cause Earth's magnetism — Earth's radius = 6400 km

Why Earth Has Magnetic Field — Possible Causes

- Masses of magnetic substances inside the Earth
- Radiation from the Sun
- Action of the Moon
- Most accepted: molten charged metallic fluid inside Earth's liquid outer core (3500 km radius) creates magnetic field
- Exact cause NOT yet fully known even today

■ Earth's Magnetism — TET Exam Trap

- Geographic North Pole $\neq$ Magnetic North Pole — they are different points on Earth
- Earth's geographic NORTH has Earth's magnetic SOUTH pole (that's why compass North points there)
- Magnetic axis is inclined 10° – 15° to geographic axis — hence compass does not point to exact geographic North
- A freely suspended magnet aligns with Earth's magnetic field (N-S direction) — used in navigation

H · USES OF MAGNETS — Daily Life & Technology

Application	How Magnet Is Used
Magnetic Compass	Directive property — finds geographic direction; used by seamen (as "direction stone"), pilots, mountaineers
Dynamo / Generator	Moving magnet near coil → generates electricity (electromagnetic induction)
Electric Bell	Electromagnet attracts hammer → strikes bell → spring pulls back → circuit breaks → cycle repeats
Electric Motor	Current-carrying coil in magnetic field rotates → converts electrical to mechanical energy
Loudspeaker / Microphone	Permanent magnet + electromagnet → converts electrical signals to sound / sound to signals
Maglev Train	Extremely powerful electromagnets — train levitates (floats) above tracks — no friction — speed up to 600 km/h
MRI (Magnetic Resonance Imaging)	Extremely powerful electromagnet scans internal organs; non-invasive medical imaging
Magnetic Crane	Powerful electromagnet lifts heavy iron/steel scrap; releases by switching off current
Hard Disk / Memory	Magnetic material stores data in computers; also used in debit/credit card magnetic strips
MICR in Banking	Magnetic Ink Character Recognition — magnets in computers read MICR numbers on cheques
Magnetic Conveyor Belt	Sorts magnetic substances from non-magnetic in industries
Screwdrivers	Tip slightly magnetised so screws remain attached while working
Alnico Cow Magnet	Swallowed by cattle — attracts sharp iron wires in digestive tract preventing internal damage
Refrigerator / Door seals	Permanent magnets in door seal keep fridge door closed tightly
Speakers / Earphones	Permanent ring magnet + voice coil electromagnet create vibration → sound

I · MAGLEV TRAIN — Magnetic Levitation Technology

Maglev	Magnetic Levitation — train floats above track using magnetic repulsion — no contact, no friction
Principle	Like poles repel → electromagnets at bottom of train + track repel → train lifts ~10 cm above track

Propulsion	Unlike poles attract → magnets on sides of track and train pull the train forward → directional control by changing current
Speed	No friction → easily attains 300 km/h; capable of up to 600 km/h
Features	No moving parts → no friction; no noise; less energy; eco-friendly
Countries	China (Shanghai Maglev: 501 km/h record), Japan (SCMaglev: 603 km/h record), South Korea (KTX: 421 km/h record)
India	Mumbai-Delhi, Mumbai-Nagpur, Chennai-Bengaluru-Mysuru routes under consideration/proposal

Maglev Speed Records		
Country / Train	Max Operating Speed	Speed Record
Japan — SCMaglev	500 km/h	603 km/h (world record)
China — Shanghai Maglev Train	350 km/h	501 km/h
South Korea — KTX	300 km/h	421 km/h

J • Key Facts — Rapid Revision (Numbers, Years & Names only)

Values, Formulas & Key Numbers

Quantity / Concept	Value / Fact	Unit / Note
Magnetic field unit	Tesla (T) — SI unit Gauss — CGS unit	1 T = 10,000 gauss
Earth's field strength	25 to 65 micro tesla at Earth's surface	μT
Magnetic axis inclination	Inclined 10°–15° to geographic (rotation) axis	Earth's magnet
Earth's core radius	~3500 km (liquid outer core)	Earth radius = 6400 km
Magnetite iron content	72.4% iron — Fe_3O_4	Strongest natural magnet
Hematite iron content	69% iron — Fe_2O_3	Common iron ore
Siderite iron content	48.2% iron — FeCO_3	Weakest iron ore
ALNICO composition	Aluminium + Nickel + Cobalt alloy	Common permanent magnet
Neodymium magnets	Strongest artificial magnets on Earth	Nd alloy
Maglev levitation height	~10 cm above track	Magnetic repulsion

Quantity / Concept	Value / Fact	Unit / Note
Japan SCMaglev record	603 km/h	Speed record
China Shanghai Maglev	501 km/h (record) 350 km/h (operating)	Speed
Credit card strip	Tiny iron-based magnetic particles (20 millionth of an inch long) in thin plastic film	Magstripe
Curie temperature	Temperature at which ferromagnetic → paramagnetic	Named after Pierre Curie
Magnetar	Most powerful magnet in universe — neutron star, 20 km diameter, 2-3 solar masses	Milky Way galaxy

Scientists, Discoveries & Years

Scientist / Event	Contribution / Fact	Year
William Gilbert	Laid foundation of magnetism; suggested Earth has giant bar magnet; first systematic research on lodestone; published "De Magnete"	1544–1603
Chinese navigators	Used magnetite (lodestone) as compass for navigation in boats during storms	~800 years ago 1200 AD
Pierre Curie	Curie temperature — ferromagnetic becomes paramagnetic on heating; named Curie temperature after him	19th century
Hans C. Oersted	Discovered magnetic effect of electric current — current produces magnetic field	1819
China (Maglev)	Shanghai Maglev operating — 350 km/h operating, 501 km/h record	2003 onward
Japan (Maglev)	SCMaglev — world speed record 603 km/h	2015

K · Glossary — Definitions

Magnet — Object that attracts iron, cobalt and nickel due to property called magnetism	Magnetism — Branch of physics dealing with properties of magnets; also the attracting property itself
Natural Magnet — Magnet found in nature — e.g., Lodestone/Magnetite (Fe $\blacksquare$ O $\blacksquare$)	Artificial Magnet — Man-made magnet — stronger than natural; made in specific shapes and sizes

Lodestone / Magnetite — Strongest natural magnet — iron oxide ore Fe_3O_4 ; used historically as compass	Pole — Region near end of magnet where attractive force is maximum; always exist in pairs (N and S)
Magnetic Meridian — Vertical plane passing through magnetic axis of Earth at a given location	Attractive Property — Magnets attract magnetic materials (Fe, Co, Ni); strongest at poles
Repulsive Property — Like poles repel each other; unlike poles attract	Directive Property — Freely suspended magnet aligns along geographic N-S direction
Magnetic Field — Space around magnet where magnetic effect is observed; unit: tesla or gauss	Tesla (T) — SI unit of magnetic field; $1 \text{ T} = 10,000 \text{ gauss}$
Gauss — CGS unit of magnetic field; $1 \text{ gauss} = 10^{-4} \text{ tesla}$	Field Lines — Curved lines showing direction and pattern of magnetic field around a magnet
Neutral Point — Point where magnetic fields of two magnets cancel; no net magnetic force	Diamagnetic Material — Aligns perpendicular to field; moves to weaker region; magnetised opposite to field; not affected by temperature
Paramagnetic Material — Aligns parallel to field; moves to stronger region; magnetised in field direction; affected by temperature	Ferromagnetic Material — Strongly aligns parallel to field; quickly moves to stronger region; strongly magnetised; becomes paramagnetic at Curie temperature
Curie Temperature — Temperature at which ferromagnetic material becomes paramagnetic	Permanent Magnet — Retains magnetism even without external field; made from ALNICO, Neodymium alloy, hardened steel
Temporary Magnet — Loses magnetism when external field removed; made from soft iron; used in electromagnets	Magnetisation — Process of making a substance a permanent or temporary magnet using external magnetic field
ALNICO — Alloy of Aluminium, Nickel and Cobalt — used for permanent magnets	Neodymium Magnet — Strongest artificial magnet on Earth — made from neodymium alloy
Compass — Instrument with magnetic needle to find geographic directions; uses directive property	Plotting Compass — Small compass with freely rotating magnetic needle; used to trace magnetic field lines
Earth's Magnetism — Earth behaves as giant magnetic dipole; magnetic axis inclined $10\text{--}15^\circ$ to geographic axis	Magnetic Declination — Angle between geographic north and magnetic north at a location
Maglev Train — Magnetic levitation train — floats above track using like-pole repulsion — no friction	Electromagnet — Magnet created by electric current through coil around iron core — temporary magnet
MRI — Magnetic Resonance Imaging — medical imaging using powerful electromagnets	Magnetar — Most powerful magnet in universe — a type of neutron star; mass 2-3 times Sun
Magneto-reception — Ability to sense Earth's magnetic field — seen in pigeons (magnetite in beaks)	Magnetic Hard material — Material requiring strong field to magnetise; retains magnetism — used for permanent magnets

Magnetic Soft material — Material easily magnetised but loses magnetism quickly — used for electromagnets	
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■ Did You Know?

- ◆ A magnetar (magnetic neutron star) in the Milky Way has a field so powerful it could absorb iron atoms from blood even 1000 km away.
- ◆ Pigeons have magnetite crystals in their beaks — they can sense Earth's magnetic field for navigation (magnetoreception).
- ◆ Credit/debit card magnetic strip contains tiny iron-based magnetic particles just 20 millionths of an inch long.
- ◆ Earth's magnet is 20 times more powerful than a fridge magnet.
- ◆ Maglev trains have no wheels, no engine, no friction — they achieve 600+ km/h floating silently above tracks.